When did living land plants arise? What their fossils require

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In one sentence: older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

The puzzle

Every forest, grassland and crop descends from a few small plants that colonised the land hundreds of millions of years ago. Those plants changed the air, the soils and the climate. Yet scientists disagree by tens of millions of years about when the ancestor of today's land plants lived.

One way to find out is a *molecular clock*: compare the DNA of living plants and work out how long their lineages have been diverging. A clock dates the shared ancestor of living species, which may be younger than the first plants on land. Clocks need fossils to set their time scale, and they lean on one kind of fossil argument in particular: a *maximum age*. A maximum says that a much older group would probably have left recognisable fossils in the rocks examined. Move the maximum and the clock's answer moves with it.

Three recent clock studies show how much rides on this. Two place the ancestor of all living land plants 515–474 million years ago (Ma), in the Cambrian or earliest Ordovician. The newest, with different fossil maximums, places it at 457–450 Ma (Morris et al. 2018; Harris et al. 2022; Hooper et al. 2026). For the ancestor of living *vascular* plants, the group with internal plumbing that includes clubmosses, ferns and seed plants, the estimates run from about 452 to 428 Ma (figure below).

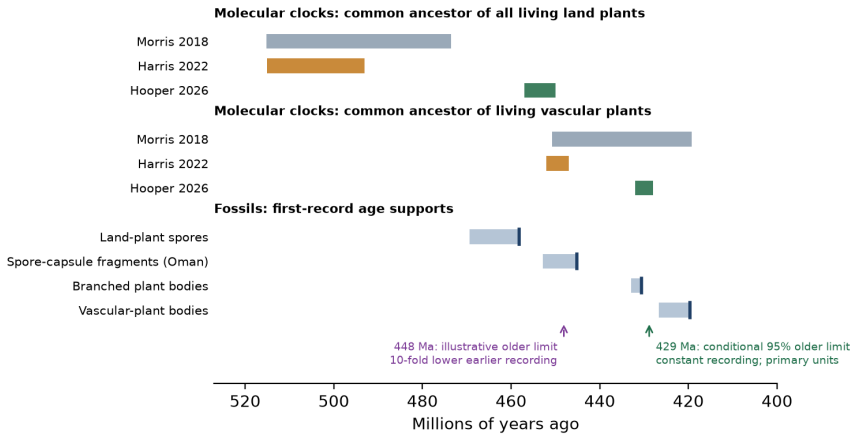


Figure. Clock estimates and first-record age ranges (bands run from the earliest possible to the earliest certain record; the two ends may be different finds). Arrows: the 95% older limit on when the vascular-plant body record began if fossils were recorded at a steady rate (green), and a retrospective scenario endpoint if fossils were recorded ten times less often before about 427 Ma (purple), chosen after seeing the data, which shows compatibility rather than a 95% limit. Both describe the fossil record; the ancestor itself can be older.

Why early plants are hard to date

The oldest traces of land plants are spores: tough, microscopic and made in huge numbers. Spores appear in the Middle Ordovician, Late Ordovician rocks in Oman preserve fragments of spore capsules, and branched plants, with stems and capsules often only millimetres long, appear in the Silurian. Each kind of fossil reaches the rocks by its own route and is found in its own way. Missing fossils in older rocks can mean the plants were absent, or that small, woodless and perhaps rare plants were seldom preserved, found and recognised. A maximum age has to argue for the first reading.

What we did

We examined the three maximum ages that divide the clocks: 515.5 and 459.1 Ma for living land plants, and 431.2 Ma for living vascular plants.

- **We traced each maximum to its original fossils**, ages and arguments.

- **We modelled spore records from Libyan boreholes.** In one North African basin, none of 28 rock samples older than about 443 Ma reports spores with a three-rayed (*trilete*) mark, while 51 of 59 younger samples do. Most of the older samples are only a little older; 10 are broadly dated and may reach back beyond 459 Ma.
- **We bounded when the vascular-plant body record began.** From how 48 dated groups of vascular-plant fossils are spread through time, a standard method gives a 95% limit on how far back the record reaches. That limit depends on how often plants left fossils that were later found and recognised, so we tested several histories of that rate.

What we found

1. **The oldest maximum (515.5 Ma) rests on a qualitative argument.** Older marine rocks preserve tough algal cysts and no land-plant spores, so its authors read the spores' absence as real. How many land spores reached those seas, and how hard anyone looked, remain unmeasured.
2. **The Libyan rocks limit sustained, well-delivered spore input.** If older spores reached the rocks as well as later ones did, a steady spore input from 459.1 Ma fits only below about a tenth of the later input (0.091). Lower input, spores that seldom reached the basin, or plants living elsewhere all fit. The model treats samples from one well as linked by a shared well effect and otherwise independent; weighting samples by well and rock unit instead gives an exploratory figure near 0.4. The result covers mainly the later Ordovician in one polar basin.
3. **Vascular plants can be older if early fossils were scarcer, or recognisable later.** At a steady recording rate, the 95% older limit on their body record is about 429 Ma, or about 432–434 Ma if the finds are counted as fewer independent places. In a retrospective scenario that halves the earlier rate, the record's start reaches only 431.0 Ma, still younger than the 431.2 Ma maximum. Dates of 444–452 Ma, as in the older clocks, fit if such fossils were recorded 8 to 12 times less often before the oldest find than after it (2.4 to 4.5 times if counted cautiously). They also fit at a steady rate if recognisable bodies appeared at least 15–23 million years after the ancestor.
4. **One find can move the limit by 20 million years.** Counting one set of tiny spore-capsule fragments from Late Ordovician Oman moves the limit for land-plant bodies back by about 20 million years.

What it means

The vascular-plant fossil record leaves room for older ancestors, provided early plants were several times scarcer in the record, or gained their recognisable features later. The Libyan spores leave room for older ancestors whose spores reached that basin sparsely or not at all. Plants that were uncommon, local, small or seldom searched for would leave exactly such a sparse record; our calculations give the size of the requirement, and its biological plausibility remains open. Clock builders can state the fossil, geographical and recording assumptions behind each maximum and test how alternative maxima change their dates. In short: older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

How this work was done

Two AI systems, Claude (Anthropic) and Codex (OpenAI), did the research with Barry Brook, who set its direction; independent reviews by GPT-6 Pro (OpenAI) shaped it. Every number traces to a file and script in the public repository.

Sources

Harris, Brogan J., James W. Clark, Dominik Schrempf, Gergely J. Szöllősi, Philip C. J. Donoghue, Alistair M. Hetherington, and Tom A. Williams. 2022. "Divergent Evolutionary Trajectories of Bryophytes and Tracheophytes from a Complex Common Ancestor of Land Plants." *Nature Ecology & Evolution* 6 (11): 1634–43. https://doi.org/10.1038/s41559-022-01885-x.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

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